

UNIT 4

DATA AND ANALYSIS

10TH COMPUTER SCIENCE FEDERAL BOARD

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AGENDA

- Introduction
 - Data Science (DS)
 - Artificial Intelligence (AI) and its scope
 - ML Models
 - Machine Learning Process Steps
 - Churn Prediction and Behavioral Segmentations
 - Data Visualization
 - Database and ML
 - Stages of DS Life Cycle
 - Case Study
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DATA SCIENCE

- Data Science is the field of collecting, cleaning, and analyzing data to find useful patterns and make better decisions.
 - It combines math, statistics, programming, and domain knowledge.
 - The available data can be structured as well as unstructured.
 - Various programming languages like Python, R studio and MATLAB can be used for data visualization.
 - Real-life examples:
 - Netflix recommends movies based on your watch history
 - Amazon suggests products you might buy
 - Doctors predict diseases from patient data
 - Banks detect fraud transactions
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DATA SCIENCE AND ITS SCOPE



Healthcare: To predict the possibilities of some disease, based on X-Rays, Ultrasounds or other analytics of patient.



Sports: If we want to predict who is going to win a cricket match, based on the previous performance of teams, data science will help us.



Finance: In the banking and finance sector, data science is helpful in fraud detection. It is also helpful in developing trading and investment strategies.

DATA SCIENCE AND ITS SCOPE



Goods transportation: It helps in route optimization and minimize delivery times. Data driven approach helps to reduce cost. It also helps in real time tracking of transportation of goods.

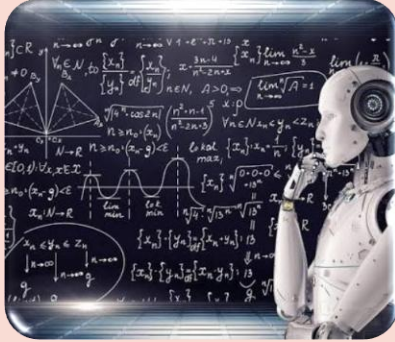


Airline: It helps in efficient route planning to reduce fuel consumption. It helps to improve services based on customer experience and feedback.



Energy and utilities: It helps to optimize energy production and distribution. Realtime monitoring is useful for load distribution and facilitates immediate response for demand fluctuations.

DATA SCIENCE AND ITS SCOPE



Education: It is useful to enhance personalized learning based on the performance of learners. It also helps to predict students' performance based on previous data.



Communication Media: It is used for content creation and distribution. Various approaches like sentiment analysis, social media monitoring and recommender systems are a few examples of scope of data science in media and communication sector.



Entertainment: Data science plays important role when you get suggestions about your favorite songs while playing online computer games.



Business: Data science helps in intelligent decision making about the business to get optimal profits. Customer insight and sentiments are also analyzed to improve the business.

ARTIFICIAL INTELLIGENCE (AI) AND ITS SCOPE

- The term artificial intelligence is not new. In 1950, a British mathematician Alan Turing proposed a Turing Test, which measured the ability of a machine to exhibit intelligent behavior. Nowadays, modern robots are supposed to be smarter if they can pass the Turing Test.
 - The term Artificial Intelligence refers to the ability of a machine to exhibit human behavior like problem solving, understanding natural language, and interacting with the environment intelligently.
 - we can discuss a few application of AI as:
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ARTIFICIAL INTELLIGENCE (AI) AND ITS SCOPE

AI Agents: Artificial intelligence provides bases to develop modern AI agents like Siri, Alexa, Google assistant, ChatGPT and Cortana.

Computer vision: It is a specialized branch of AI that teaches computers to draw meaningful results from digital images, videos and other visuals.

Healthcare: Like data science, artificial intelligence has played equally well in the field of healthcare. The examples are personalized treatment recommendations based on medical imaging.

Robotics: Whether you watch some latest robot, talking and cooking like a human being or you get a small robot to clean your house, all this technology is due to significant advancement of artificial intelligence.

Natural Language Processing (NLP): AI enabled the machines to understand and respond to the natural human languages. ChatGPT and chatbots are examples of it.
Now you can switch ON your air conditioner with voice command rather than picking up the remote controller.

ML MODELS: SUPERVISED LEARNING MODEL

- In this learning model a set of **labeled data** is provided to the machine/computer. The labeled data means that questions with their correct answers are provided to the machine. After reading/learning a lot of questions/answers the machine is trained. Now, only new questions without correct answers are provided to that machine and it provides us with the possible correct answers.
 - For example, if we provide the features (labels) of types of fruits i.e. shape, color and texture and give their names (answers) as well. After the machine is trained when we will give the features of a new fruit the supervised learning model will predict/provide the name of that fruit.
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ML MODELS: UNSUPERVISED LEARNING MODEL

- In unsupervised learning model, unlabeled data is provided to the machine/computer. The algorithm identifies the pattern in the data and divide it into groups based on similarities and differences. When the training is completed, the computer can identify new data and put it in the relevant category.
 - For example, if we provide various types of news to a computer then it will start dividing them into various groups, based on keywords. Suppose from a huge dataset the computer divides the news into various categories i.e. national politics, international politics, entertainment, medical, and technological news. When we give a new piece of information "Scientists successfully, saved the cancer patients at last stage" to the computer, it will categorize it into medical news.
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ML MODELS: REINFORCEMENT LEARNING MODEL

- In reinforcement learning the machines are trained with the help of positive or negative feedback. To explain reinforcement learning we will use the following key terms:
 - **Agent** (machine/computer)
 - **Environment** (where agent takes actions)
 - **Actions** (moves made by agent)
 - **Rewards** (points for helpful actions)
 - **Penalties** (points deducted for unhelpful actions)
 - The agent is provided with an environment where it must take certain actions, if the action is helpful towards the desired solution the agent will be rewarded with some points through feedback. If the actions are not helpful towards the desired solution the agent will be penalized (punished) by deducting the points. The agent improves its actions based on the feedback. This model is used to get optimal results.
 - Examples: Self-Driving Cars, AI Competition, Robots, Trading Algorithms
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MACHINE LEARNING PROCESS STEPS

Step 1: The preprocessed, cleaned data is split into **Training data** and **Testing data**.



Step 2: A suitable algorithm is selected to train the machine based on problem and data.



Step 3: The model is trained on the training set; it learns the pattern from data and develop a decision-making ability.



Step 4: After the model is trained, test data is provided to the model and results are compared with actual values. If needed, model is improved by changing the algorithm or by adjusting other necessary parameters.

DIFFERENCES (AI, MACHINE LEARNING, DATA SCIENCE)

Aspect	Artificial Intelligence (AI)	Machine Learning (ML)	Data Science (DS)
Techniques	Neural networks, expert systems, natural language processing	Supervised, unsupervised, reinforcement, and deep learning	Statistical analysis, data mining, data visualization
Applications	Speech & image recognition, autonomous vehicles, chatbots, virtual assistants	Recommendation systems, fraud detection, predictive analysis, spam filtering, personalized marketing	Business intelligence, market analysis, healthcare analytics, risk management, scientific research
Output	Mimics human behavior	Prediction and classification	Insights, reports, and visualization

CHURN PREDICTION AND BEHAVIORAL SEGMENTATION

- **Churn Prediction** is the technique of finding out the **possibility of a customer stopping** the use of a company's services or products. It is used in **business and marketing**. For example, a telecom company receiving frequent complaints can predict that those customers might **switch to another company**. Keeping existing customers is often **more beneficial than getting new ones**.
 - *Example:* A clothing store collects data on **purchase frequency, brand loyalty, and feedback**. After analysis, if some customers are visiting less often, the store can send **personalized offers** to retain them.
 - **Behavioral Segmentation** is the process of **dividing customers into groups based on their behavior**. It helps businesses design **precise marketing strategies** and meet the **specific needs** of each segment. Key aspects: **purchase frequency, brand loyalty, usage occasions, and response to market stimuli**.
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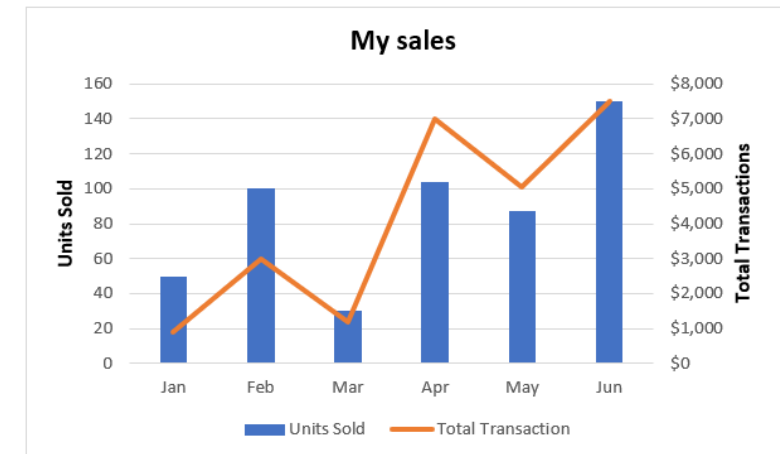
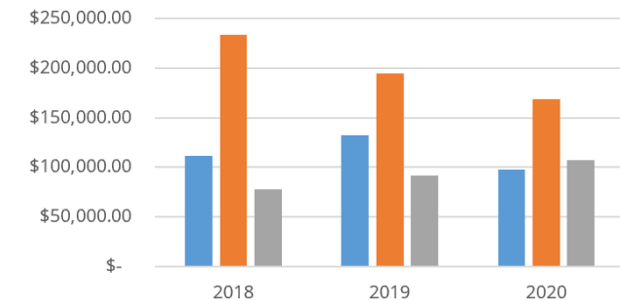
DATA VISUALIZATION

- **Data Visualization** is the process of creating **graphical representations** of data using **charts, graphs, and maps**. It transforms **raw data into meaningful, insightful information**, making complex data **simple to understand** and helping in **decision-making** across science, arts, and daily life.
 - Interactive visualization techniques help get **optimal results** from available data. There are **various techniques and types** that help extract meaningful information for **business and service improvement**.
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METHODS OF DATA VISUALIZATION

- Methods and types of data visualization are interrelated terms. However, methods of data visualization are the techniques to prepare data for visualization. These methods are explained briefly as follows:
- **Bar Chart:** This method of data visualization is used to show comparison among discrete values. For example, we can use Bar Chart to visualize marks of students in various subjects or sales data of various products in a company.
- **Line Chart:** It is used to display trends of data over short or long period of time. For example, gold prices over the last five or ten years can be displayed by using a Line Chart. Similarly, stock prices, or change in temperature over a year can be presented by using a Line Chart.

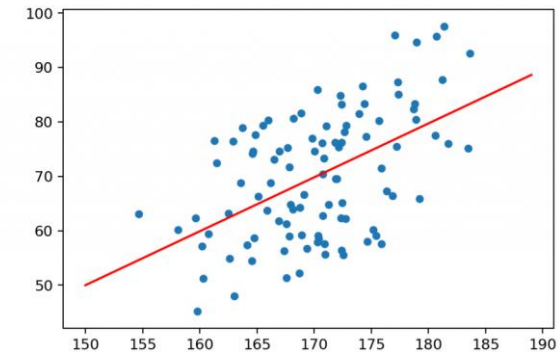
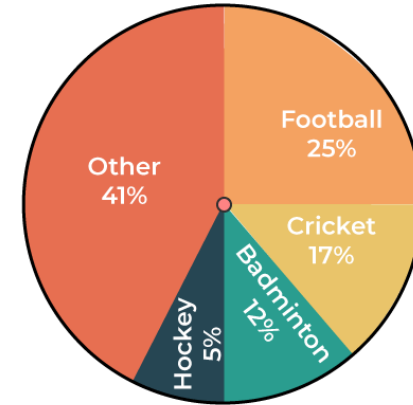
Bar Chart (Data Visualization)



METHODS OF DATA VISUALIZATION

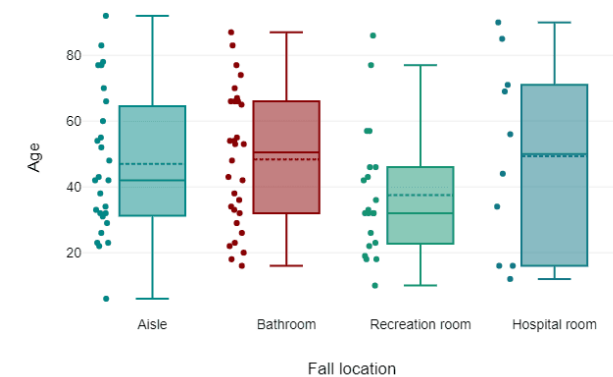
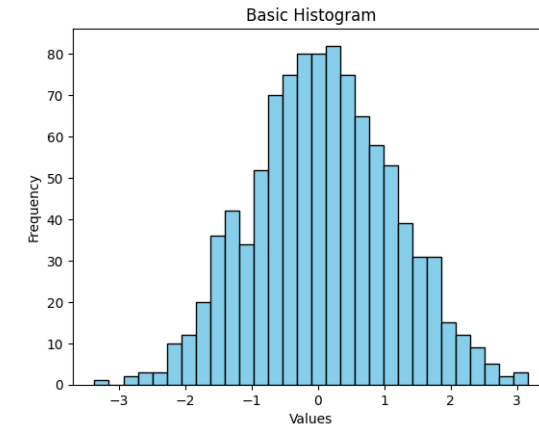
- **Pie chart:** it is used to represent proportions of various categories of data in a whole. For example, market share of various companies.
- **Scatter Plot:** it is used to display the relationship between numerical variables. For example: correlation between height and weight of a group people.

Number of Students



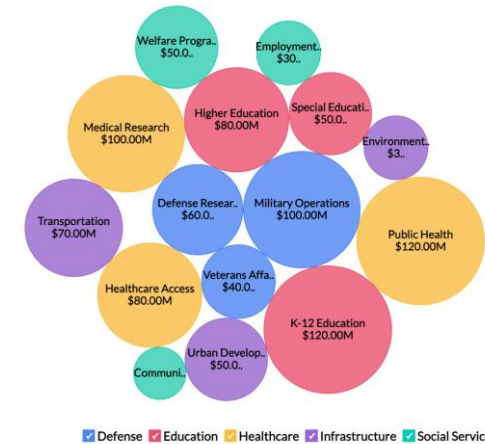
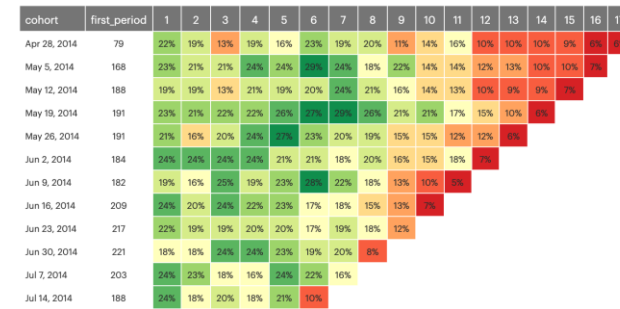
METHODS OF DATA VISUALIZATION

- **Histogram:** it is used to show the distribution of a single numerical value. For example, distribution of score in a class.
- **Box-plot (Box-and-Whisker Plot):** it is used to summarize the distribution of data with the help of minimum, first quartile, median, third quartile and maximum data. For example, analysis of salaries of various employees in company.



METHODS OF DATA VISUALIZATION

- **Heatmap:** it is used to represent data in matrix form. Data values are shown by using color intensity. For example, images received from the space to analyze weather forecast. Another example is of climate change, or measurement of different of flowers.
- **Bubble Chart:** it is similar to scatter plot, but is has a third variable which is represented by size of bubble. For example, points in a 2D space where area of each point is proportional to the third variable.



TYPES OF DATA VISUALIZATION

- There are different ways to represent data graphically to make it easier to understand. Some common types of data visualization are as follows:
 - **Quantitative Visualization** – Represents **numerical/measurable data**.
Example: Bar chart showing monthly sales figures.
 - **Categorical Visualization** – Represents data in **distinct categories**, showing proportions or parts of a whole. Ideal for **nominal or ordinal data**.
Example: Pie chart showing market share of companies.
 - **Temporal Visualization** – Displays data that **changes over time** (time-series data). Usually shown with **line graphs**.
Example: Line graph of temperature over a year.
 - **Spatial Visualization** – Represents data related to **physical locations or spaces** (geographic data).
Example: Heatmap showing population density across regions.
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TYPES OF DATA VISUALIZATION

- **Multivariate Visualization** – Represents data with **more than two variables**.
Tools: Scatter plots, heatmaps.
Example: Scatter plot showing income, age, and spending habits.
 - **Interactive Visualization** – Allows users to **interact with data** through dashboards and filters.
Tools: Power BI, Tableau.
Example: A dashboard where you filter sales by region and year.
 - **Statistical Visualization** – Shows **statistical properties** like distribution or correlation.
Tools: Histograms, box plots, scatter plots.
Example: A histogram showing the distribution of exam scores.
 - **Information Visualization** – Presents **complex or abstract data** in an easier way.
Tools: Network diagrams, tree maps, word clouds.
Example: A network diagram showing friendships in a social network.
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STAGES OF THE DATA SCIENCE LIFE CYCLE

- Data science life cycle is an iterative process. It has various stages; each has certain steps to be taken to move forward. It requires a structured approach to solve data driven problems. The stages of data science life cycles are as follows:
 1. **Problem Definition:** Problem Definition – Define the business problem with clear objectives to achieve. Set KPIs (measurable targets) to track success. Output: a well-defined problem or hypothesis. For example, increasing sales by 15% in two months can be set as a KPI of some business problem. Another example of KPI is improve the speed of website loading by 20% in 30 days.
 2. **Data Collection:** Data Collection – Gather data from reliable sources using observations, surveys, experiments, or APIs. Data must be relevant and accurate.
 3. **Data Cleaning (Preprocessing):**– Fix errors, remove duplicates, fill missing values with estimated values, and organize data into a workable format. New features may be created from existing data. For example, if we have dates, we might create a new column for the day of the week. Clean and well-organized data is essential for making good predictions later on.
 4. **Data Analysis:**
After cleaning the data, we use statistics or machine learning to find patterns and trends. Visualization tools (graphs, charts) help understand the data. For example, we might want to know if a certain factor is causing an outcome, like is the weather affecting sales of certain products? At this stage, visualization tools like graphs or charts can be used to understand what's happening in the data.
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STAGES OF THE DATA SCIENCE LIFE CYCLE

5. **Data Modeling:** Data Modeling – Organize data into a structured model showing relationships between entities. Test it with historical data.
 6. **Model Evaluation:**
Model Evaluation – Measure the model's accuracy and reliability against the goals. Ensure security and privacy. If it fails, go back and improve.
 7. **Model Deployment** – Set up the model in a website, app, or database for real use. May be done in small steps or all at once.
 8. **Maintenance and Monitoring** – Continuously monitor accuracy, speed, and reliability. Use user feedback to improve the model as conditions change.
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